

QUALITY360 NEXT

PRODUCT REQUIREMENTS DOCUMENT (PRD)

PHASE 1 – INCOMING MATERIAL INSPECTION

1. PRODUCT VISION

Quality360 is a modern Quality Management and Laboratory Information Management platform for Metals, Mining, Smelters, Carbon Plants, Aluminum, Steel and Manufacturing industries.

The first implemented module shall be:

Incoming Material Inspection

The implementation must not be a throwaway demo.

The architecture, UX patterns, workflow engine, audit framework, notification framework and integration framework must be reusable for future modules:

- Heat Chemistry
- Casting Quality
- Mechanical Testing
- MTC & Dispatch
- Instrument Integration Hub
- Reports & Analytics

The objective is to create a production-ready application using mock data.

2. BUSINESS SCENARIO

A truck carrying Aluminum Scrap arrives at the plant.

The material cannot be consumed until:

1. Receipt is recorded
2. Sample is collected
3. Required tests are assigned
4. Results are captured
5. Specifications are validated

6. Quality team reviews findings
7. Material is approved, held or rejected

The entire process should be manageable through a single workspace.

3. DESIGN PRINCIPLES

The application should feel like:

- Modern Manufacturing Control Tower
- Enterprise SaaS Product
- Production Application

The application should NOT feel like:

- Traditional LIMS
- SAP transaction screen
- Excel replacement
- Wireframe
- Prototype

Users should understand the current situation within 3–5 seconds.

4. TECHNOLOGY STACK

Frontend:

- Next.js
- TypeScript
- TailwindCSS
- shadcn/ui
- React Query
- Recharts

Backend:

- Python FastAPI
- PostgreSQL
- SQLAlchemy
- Alembic

Future Services:

- Instrument Integration Hub (Python)

- Quality Insights Service
 - Reporting Service
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5. USER ROLES

Stores Executive

- Create Receipt
- Edit Receipt

Sampler

- Create Sample
- Recollect Sample

Lab Analyst

- Enter Results
- Upload Results
- Simulate Instrument Imports

QA Engineer

- Review Results
- Hold
- Recommend

QA Manager

- Approve
- Reject
- Override

Viewer

- Read Only
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6. NAVIGATION

Quality Operations Dashboard

Incoming Material Inspection Instrument Integrations Reports & Analytics Master Data Settings

7. PAGE 1 – INSPECTION QUEUE

Purpose: Display all incoming lots awaiting action.

Features:

- Create New Receipt
- Open Inspection
- Clone Receipt
- Cancel Receipt
- Search
- Filters

Columns:

Lot Number Supplier Material Quantity Status Risk Level Assigned User Action

Status Values:

Pending Sampling Pending Testing Pending Review Approved Rejected On Hold

8. PAGE 2 – INCOMING MATERIAL INSPECTION WORKBENCH

This is the primary screen.

All users interact here.

Layout:

1. Header
 2. Workflow Progress
 3. Material Overview
 4. Sample Section
 5. Test Results Workspace
 6. Instrument Activity Feed
 7. Quality Insights
 8. Approval Center
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9. HEADER

Display:

Lot Number Supplier Material Quantity Status Risk Level

Actions:

Edit Refresh View History

10. WORKFLOW TIMELINE

Stages:

Receipt Sample Testing Validation Review Release

Visual progress required.

11. MATERIAL OVERVIEW

Fields:

Supplier Material Lot Number Quantity Vehicle Number PO Number Receipt Date

Actions:

Edit View Supplier History

12. SAMPLE MANAGEMENT

Fields:

Sample ID Collection Date Collected By Status

Actions:

Create Sample Recollect Sample

Sample Generation should show processing messages.

Example:

Generating Sample... Assigning Workflow... Sample Created Successfully

13. TEST RESULTS WORKSPACE

Supported Result Sources:

1. Instrument Import
2. Manual Entry
3. File Upload

Tests:

XRF OES Moisture Carbon Sulphur Oxygen Custom Tests

Columns:

Test Source Status Result Action

14. INSTRUMENT SIMULATION

Supported Demo Instruments:

Thermo OES-01 Panalytical XRF-01 LECO CS-01 Moisture-01

When Import Result is clicked:

Show realistic sequence:

Connecting to Instrument... Verifying Communication... Reading Sample... Parsing Result File... Validating Result Structure... Import Successful

Duration: 4-5 seconds

Populate mock results afterwards.

15. MANUAL RESULT ENTRY

Mandatory reason:

Instrument Offline Integration Unavailable External Lab Emergency Entry Other

Audit Trail required.

16. FILE UPLOAD

Supported:

PDF CSV Excel Images

Demo Flow:

File Uploaded Extracting Values... Validation Complete

Populate mock values.

17. INSTRUMENT ACTIVITY FEED

Display live activity.

Examples:

Sample Created Import Started Results Captured Validation Complete Approval Pending

Newest first.

18. QUALITY INSIGHTS PANEL

Do not use the term AI.

Display:

Recommended Action Risk Level Supplier Health Tests Completed Key Observations Historical Comparison

Example:

Recommended Action: APPROVE

Risk: LOW

Supplier Health: 87/100

Observations:

- Chemistry compliant
 - No unusual variation
 - Supplier accepted in previous deliveries
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19. APPROVAL CENTER

Actions:

Approve Hold Reject

Hold and Reject require reason.

Messages:

Material Approved Successfully

Material Placed On Hold

Material Rejected Successfully

20. NOTIFICATIONS

Every user action must produce feedback.

Examples:

Receipt Created Sample Created Result Imported Validation Complete Approval Complete

Use toast notifications.

21. AUDIT TRAIL

Every action logged.

Store:

User Timestamp Action Previous Value New Value

Viewable from workbench.

22. CRUD REQUIREMENTS

Receipts Create Read Update Cancel Clone

Samples Create Read Update Delete Recollect

Results Import Create Update Delete Retest

Approvals Approve Hold Reject Override

23. DEMO DATA

Suppliers: ABC Metals Premium Scrap Resources Global Alloy Traders

Materials: Aluminum Scrap Primary Aluminum Silicon Metal Calcined Coke

Instruments: Thermo OES-01 Panalytical XRF-01 LECO CS-01 Moisture-01

24. FUTURE REUSE REQUIREMENT

The same framework must support:

Heat Chemistry Casting Quality Mechanical Testing MTC & Dispatch

without redesigning architecture or UX.

This is mandatory.